This section will cover the fundamental aspects of properties and profiles for cloud native applications, including how to define custom properties and when to use profiles.

4.1.1 Properties: Key/value pairs for configuration

Properties are key/value pairs supported in Java as first-class citizens with `java.util .Properties`. They play an essential role in many applications for storing configuration parameters outside of the compiled Java code. Spring Boot automatically loads them from different sources. When the same property is defined in multiple sources, there are rules that determine which one takes precedence. For example, if you specify a value for the `server.port` property both in a property file and a command-line argument, the latter takes precedence over the former. The following is a prioritized list of some of the most common property sources, starting with the highest priority:

1. `@TestPropertySource` annotations on test classes
2. Command-line arguments
3. JVM System properties from `System.getProperties()`
4. OS environment variables from `System.getenv()`
5. Configuration data files
6. `@PropertySource` annotations on `@Configuration` classes
7. Default properties from `SpringApplication.setDefaultProperties`